

Bitwise differential trail through **Blink-64** found by SAT

CiVerLy

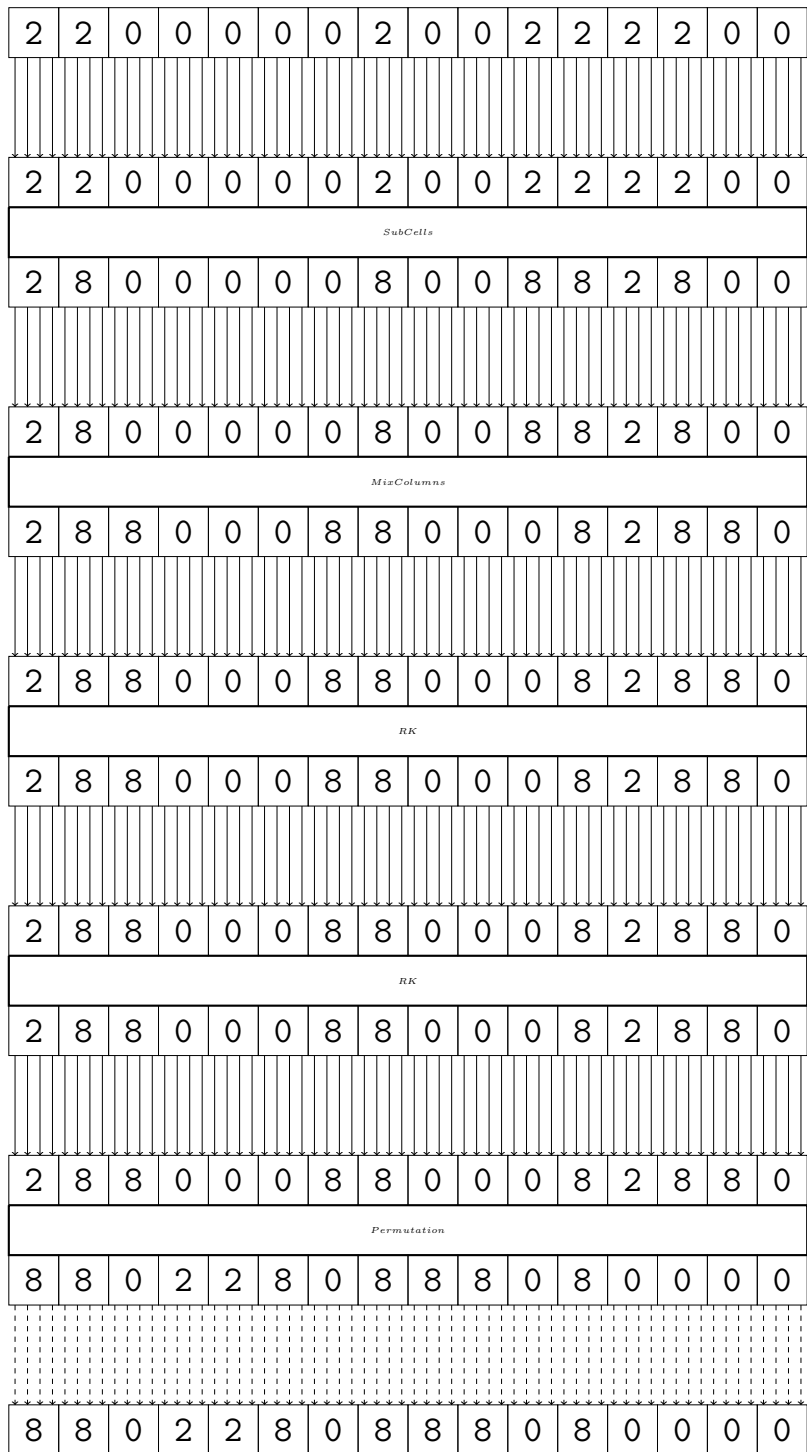
July 24, 2026

1 Blink-64

Maximal differential probability: 2^{-46}

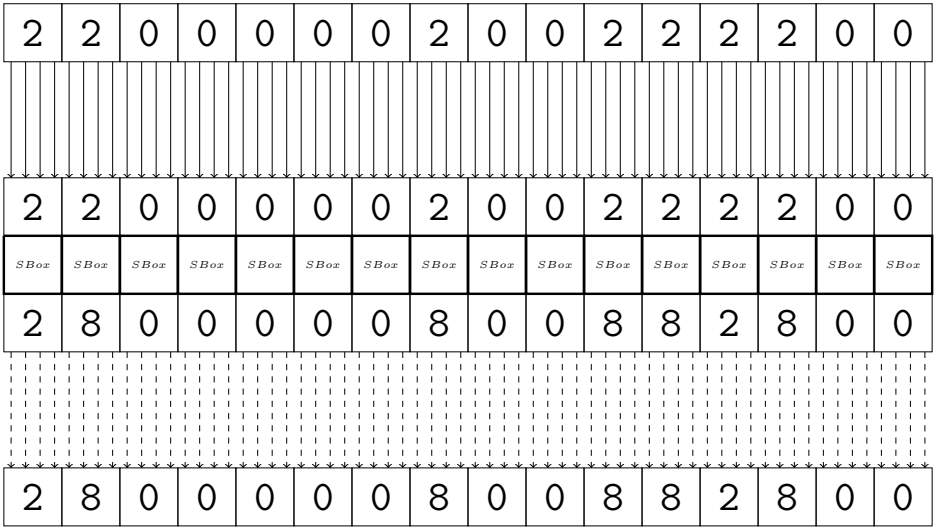
2	2	0	0	0	0	0	2	0	0	2	2	2	2	0	0
2	2	0	0	0	0	0	2	0	0	2	2	2	2	0	0
BK															
2	2	0	0	0	0	0	2	0	0	2	2	2	2	0	0
2	2	0	0	0	0	0	2	0	0	2	2	2	2	0	0
FwdRound															
8	8	0	2	2	8	0	8	8	8	0	8	0	0	0	0
8	8	0	2	2	8	0	8	8	8	0	8	0	0	0	0
FwdRound															
0	0	0	8	0	0	0	5	0	0	0	0	0	0	0	8
0	0	0	8	0	0	0	5	0	0	0	0	0	0	0	8
SubCells															
0	0	0	8	0	0	0	8	0	0	0	0	0	0	0	8
0	0	0	8	0	0	0	8	0	0	0	0	0	0	0	8
MixColumns															
0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
BK															
0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
Permutation															
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0
FwdRound															
0	2	0	0	0	0	0	2	0	0	0	0	2	0	0	0
0	2	0	0	0	0	0	2	0	0	0	0	2	0	0	0
FwdRound															
0	0	2	0	2	0	2	2	0	2	2	0	2	2	0	2
0	0	2	0	2	0	2	2	0	2	2	0	2	2	0	2

2 FwdRound

Maximal differential probability: $2^{-14.0}$ 

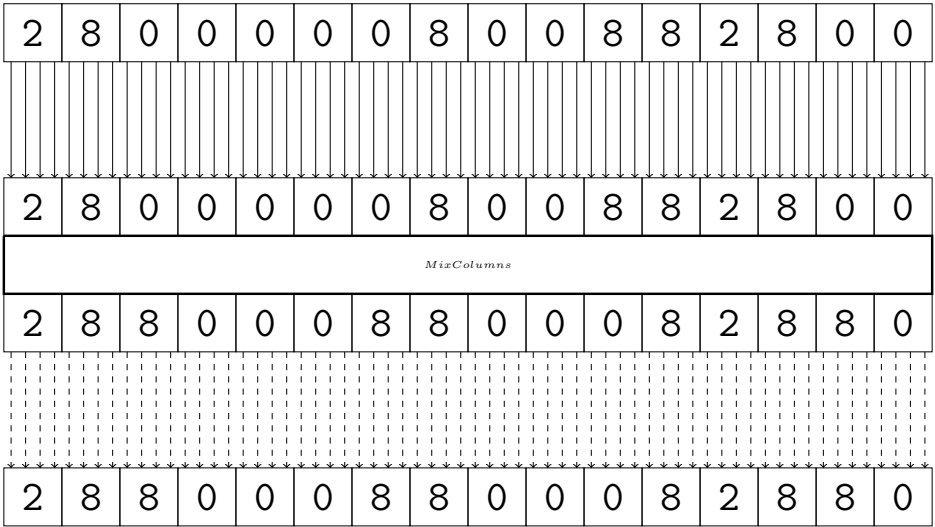
3 SubCells

Maximal differential probability: $2^{-14.0}$



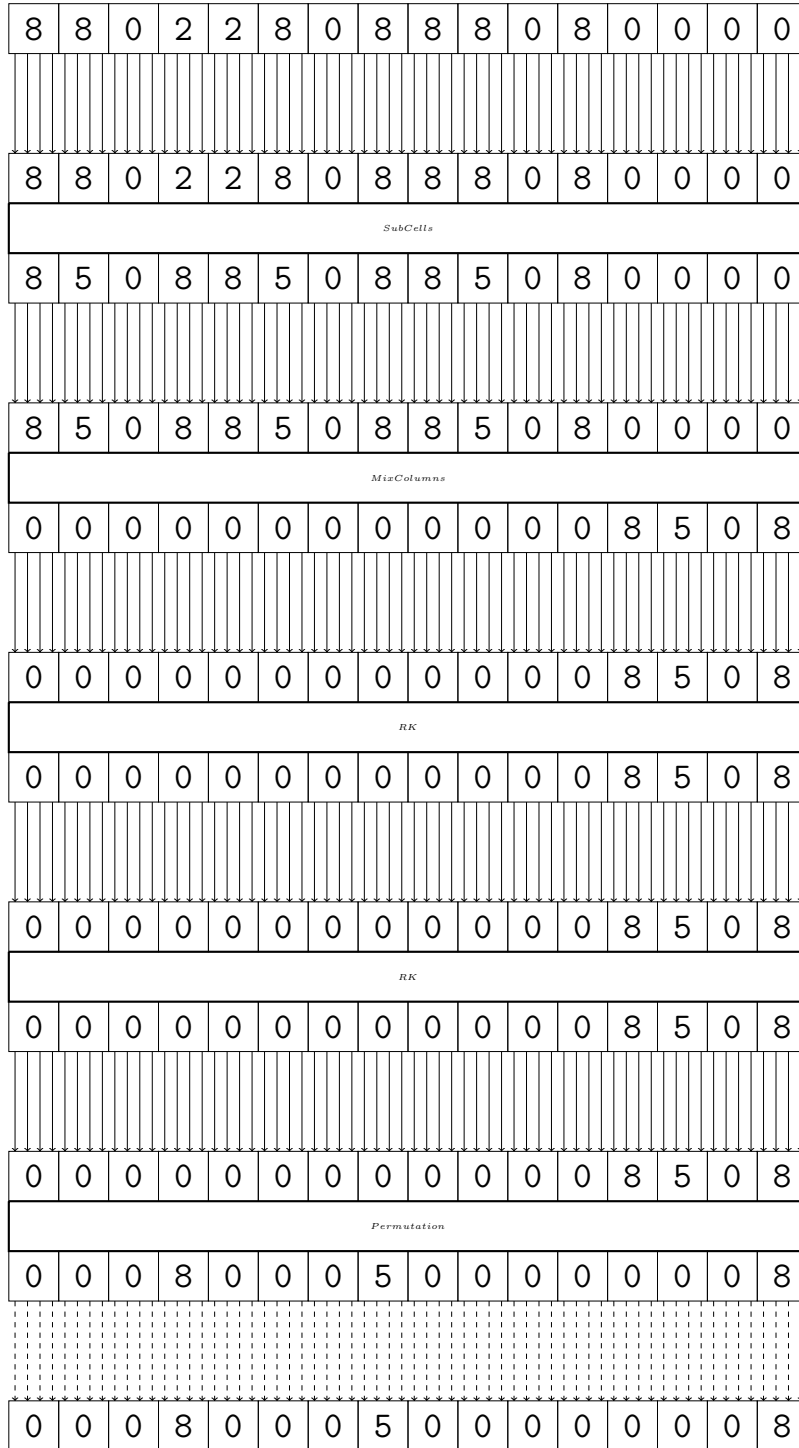
4 MixColumns

Maximal differential probability: $2^{-0.0}$



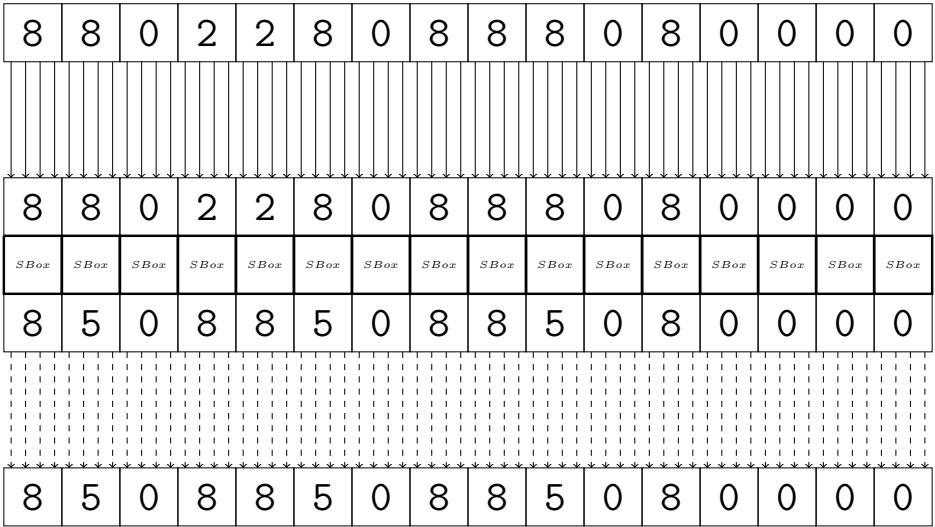
5 FwdRound

Maximal differential probability: $2^{-18.0}$



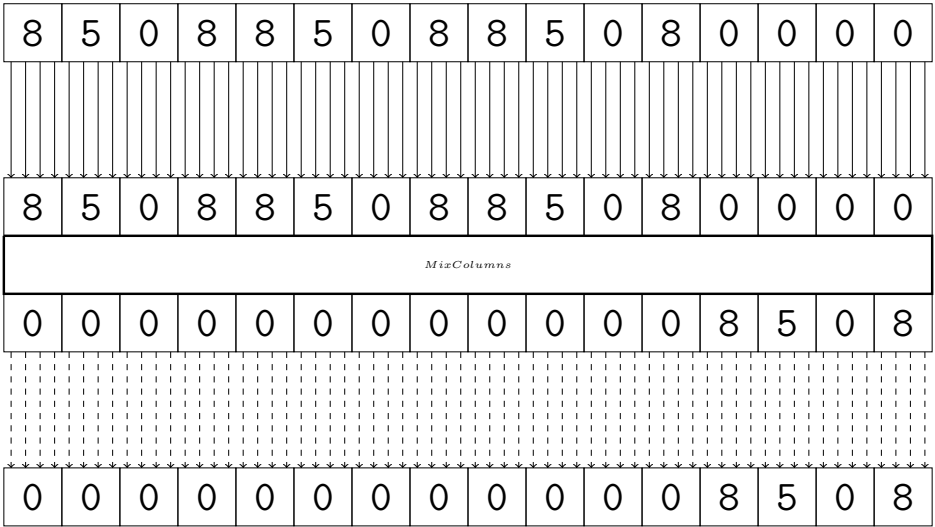
6 SubCells

Maximal differential probability: $2^{-18.0}$



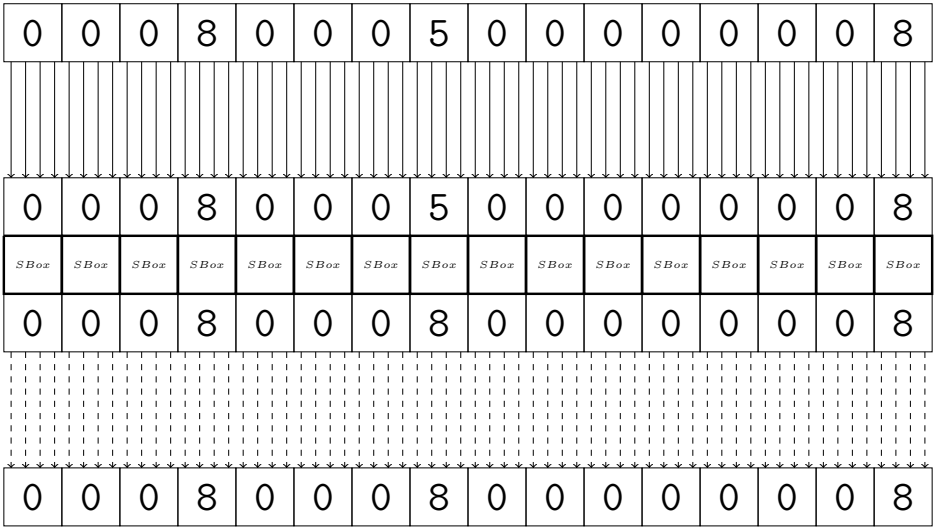
7 MixColumns

Maximal differential probability: $2^{-0.0}$



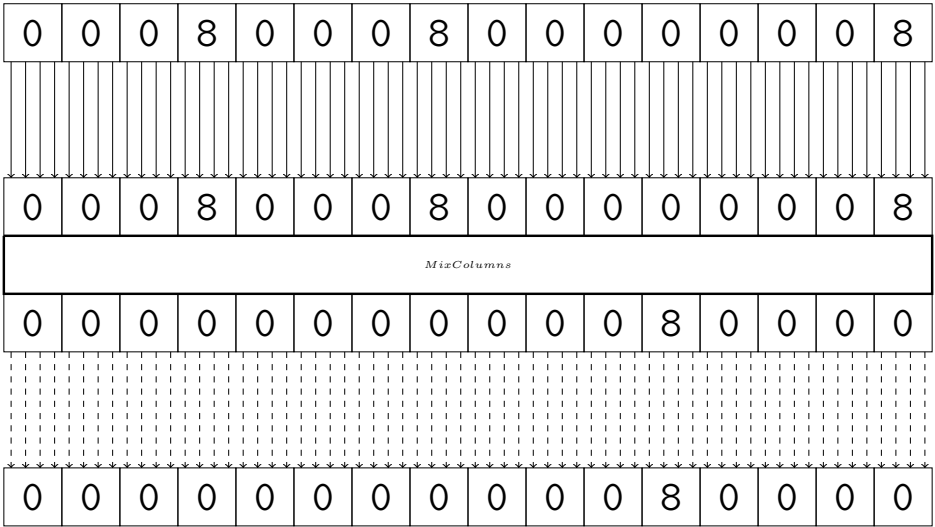
8 SubCells

Maximal differential probability: $2^{-6.0}$



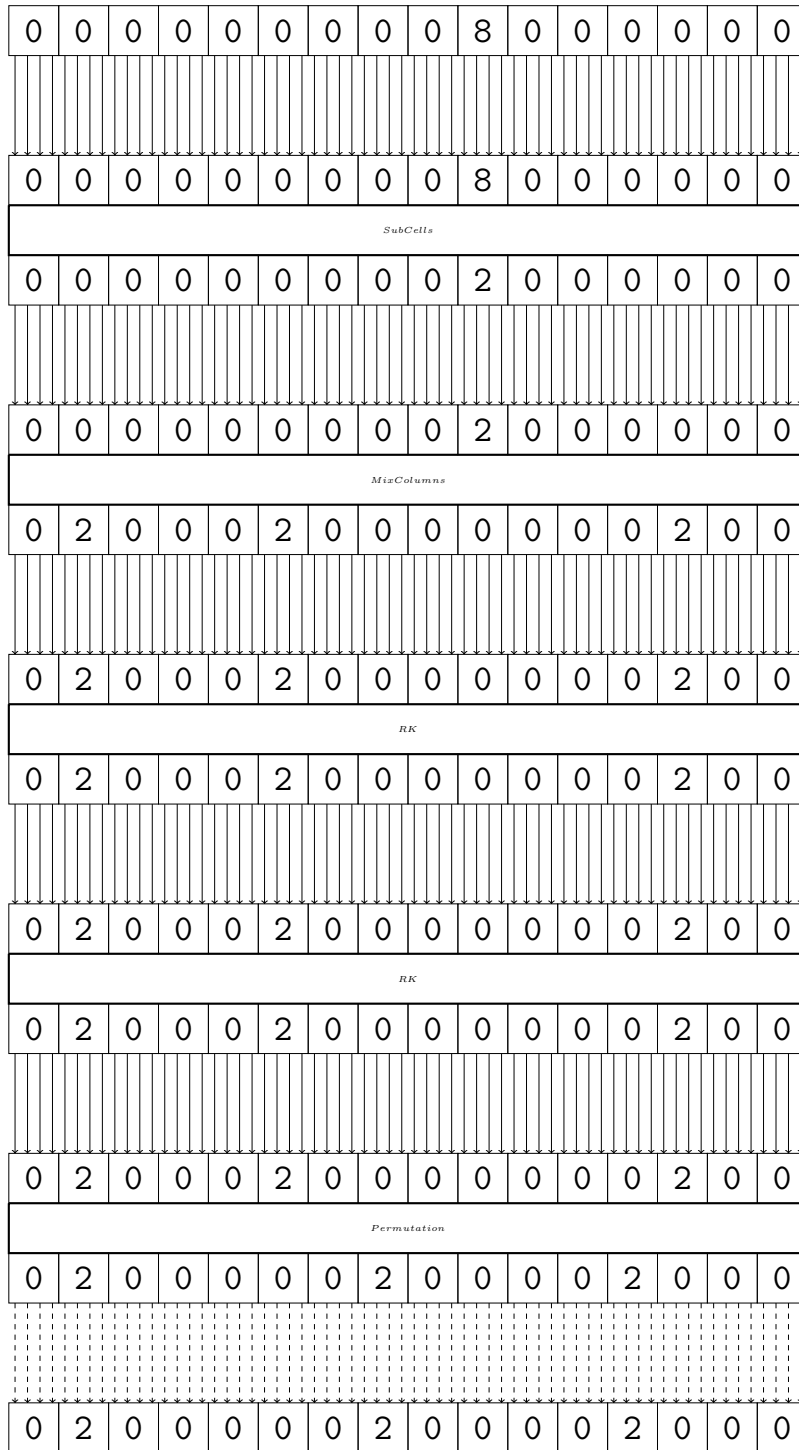
9 MixColumns

Maximal differential probability: $2^{-0.0}$



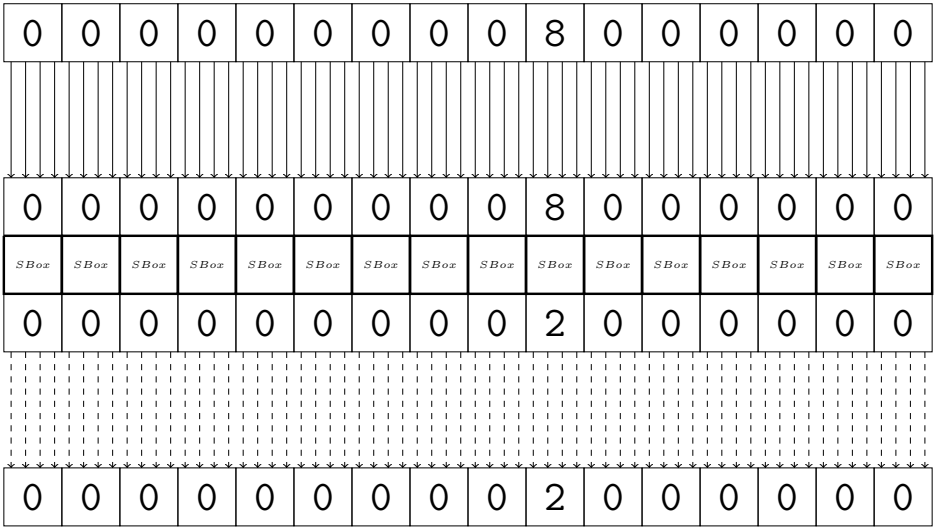
10 FwdRound

Maximal differential probability: $2^{-2.0}$



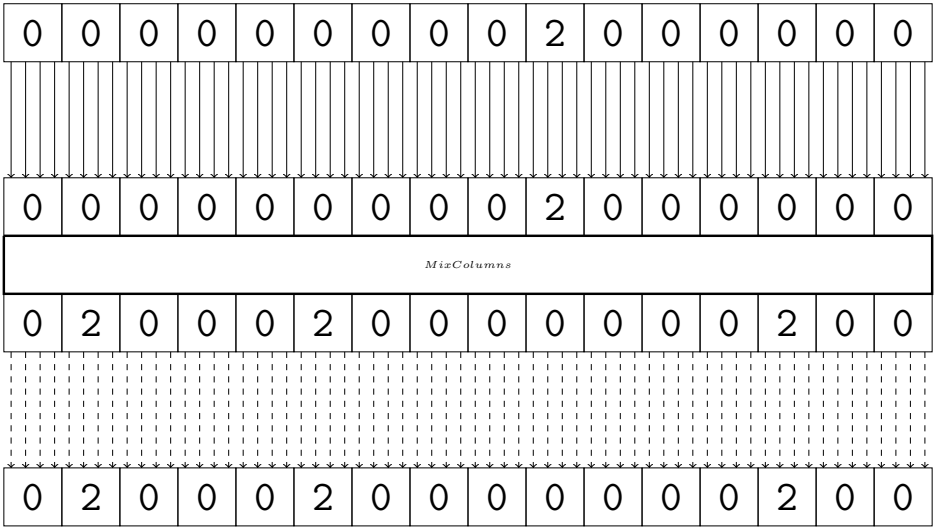
11 SubCells

Maximal differential probability: $2^{-2.0}$



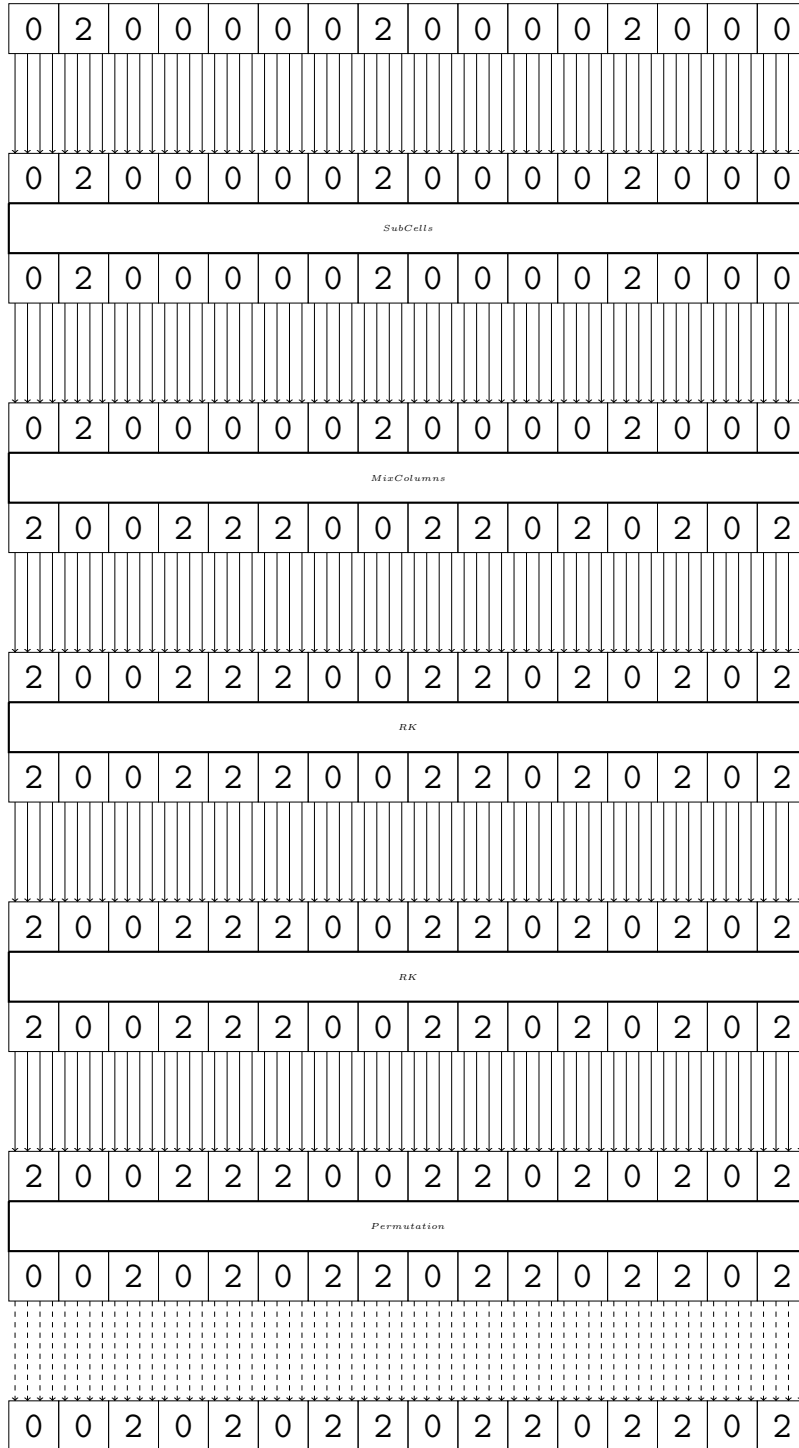
12 MixColumns

Maximal differential probability: $2^{-0.0}$



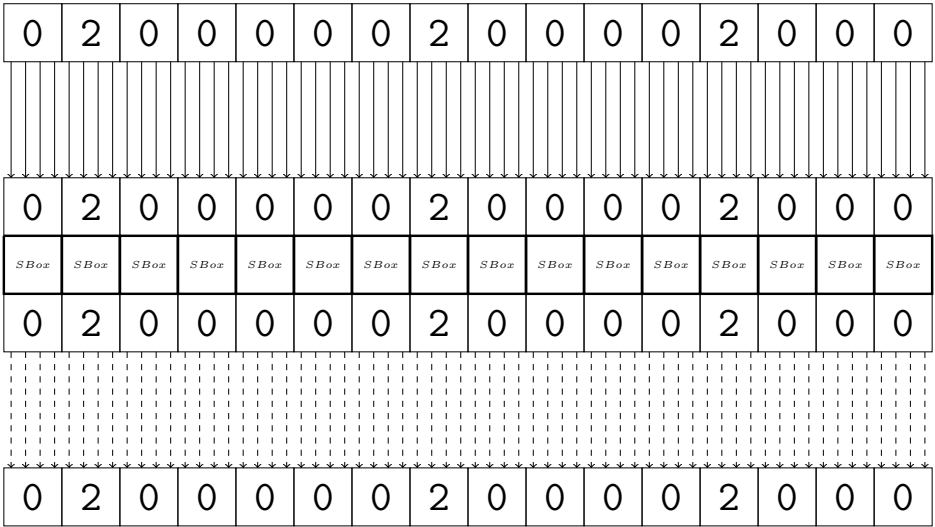
13 FwdRound

Maximal differential probability: $2^{-6.0}$



14 SubCells

Maximal differential probability: $2^{-6.0}$



15 MixColumns

Maximal differential probability: $2^{-0.0}$

